

Revision 2021

Course Code: 2069

Basic Textile Mechanics Lab

Diploma in Textile Technology | Semester 2

[Official Syllabus PDF Source](#)

Course Overview

Category	Engineering Science
Periods/Week	3 (Practical)
Total Periods	45
Credits	No Credit

Course Objectives

- Impart basic knowledge of mechanical elements like shafts, pulleys, axles, keys, couplings, and bearings.
- Understand drive transmission concepts and speed calculations.
- Study applications of gears, belts, cams, and tappets in textile machinery.

Course Outcomes (CO)

CO No.	Description	Hours	Cognitive Level
CO1	Identify mechanical elements in textile machines and explain functions.	9	Understanding
CO2	Illustrate and compare methods of drive transmission.	12	Understanding
CO3	Explain gear types, applications, and calculate speed from gearing diagrams.	13	Applying
CO4	Discuss basic aspects of cam, tappet, and variable speed drives.	8	Understanding

Module 1: Mechanical Elements

Duration: 9 Hours | CO1

M1.01 & M1.02: Shaft, Axle, Pulley, Keys, and Coupling

1. Shaft vs. Axle

Shaft: A rotating machine element used to transmit power from one place to another. In textile machines, main shafts drive multiple units.

Axle: A stationary or rotating member that supports rotating wheels, pulleys, or gears but does *not* transmit torque/power.

Malayalam Note: Shaft ഷാഫ്റ്റ് റോട്ടേറ്റിംഗ് മെക്കാനിക്കൽ ഘടകമാണ്. ഇത് ഒരു സ്ഥലത്ത് നിന്ന് മറ്റൊരു സ്ഥലത്ത് പവർ സംവഹിക്കാൻ ഉപയോഗിക്കുന്നു. (ഉദാഹരണം: ഷാഫ്റ്റ് റോട്ടേറ്റിംഗ് മെക്കാനിക്കൽ ഘടകമാണ്). Axle അക്സൽ ഷാഫ്റ്റ് അല്ല. അത് റോട്ടേറ്റിംഗ് മെക്കാനിക്കൽ ഘടകമാണ്, പക്ഷെ ടോർക്ക്/പവർ സംവഹിക്കുന്നില്ല.

2. Pulleys

Pulleys are used to transmit power using belts. Common types in textile mills:

- **Flat Pulley:** Used with flat belts for long-distance transmission.
- **V-Pulley:** Used for high torque at short distances.
- **Fast and Loose Pulley:** Used to start or stop a machine without stopping the main drive shaft.

3. Keys and Couplings

Keys: Temporary fasteners used to connect a shaft to a pulley or gear to prevent relative motion. Types: Sunk key, Woodruff key, Feather key.

Couplings: Used to connect two shafts together at their ends.

- *Rigid Couplings:* Used when shafts are perfectly aligned.
- *Flexible Couplings:* Used when there is slight misalignment (e.g., Universal coupling).

Module 2: Drive Transmission

Duration: 12 Hours | CO2

M2.01: Flexible vs. Rigid Drives

Feature	Flexible Drive (Belts/Chains)	Rigid Drive (Gears)
Distance	Suitable for long distances.	Suitable for short distances.
Shock Absorption	Absorbs shocks and vibrations.	Transmits shocks directly.
Slip	Possibility of slip (except timing belts).	No slip (Positive drive).
Maintenance	Requires periodic tensioning.	Requires lubrication.

M2.02 & M2.03: Belt and Chain Drives

Belt Drives

In textile machines like the Carding machine or Spinning frame, belts are vital.

Open Belt Drive: Shafts rotate in the same direction. **Cross Belt Drive:** Shafts rotate in opposite directions.

Roller Chain Drive

Used where a positive drive is needed without slip, but the distance is too large for gears. Commonly seen in the drive to the lattice in blowroom machinery.

Malayalam Note: റബ്ബർ ബെൽ ഡ്രൈവ് (Slip) പോസിറ്റീവ് ഡ്രൈവ് അല്ല. റബ്ബർ ബെൽ ഡ്രൈവ് പോസിറ്റീവ് ഡ്രൈവ് അല്ല, റബ്ബർ ബെൽ ഡ്രൈവ് പോസിറ്റീവ് ഡ്രൈവ് അല്ല.

Module 3: Gears and Calculations

Duration: 13 Hours | CO3

M3.01 & M3.02: Types of Gears

- **Spur Gear:** Teeth are parallel to the axis. Used for parallel shafts.
- **Bevel Gear:** Used for shafts intersecting at 90 degrees (e.g., Differential motion in Roving frame).
- **Worm and Worm Wheel:** Used for large speed reduction in a small space. The worm is like a screw.
- **Pawl and Ratchet:** Allows motion in only one direction. Used in take-up motions of looms.

M3.03: Gear Trains and Speed Calculations

Simple Gear Train: Each shaft carries only one gear.

Compound Gear Train: At least one shaft carries two gears. This is used to achieve high velocity ratios in limited space.

Velocity Ratio Formula:

For two gears 1 (Driver) and 2 (Driven):

$$N_2 / N_1 = T_1 / T_2$$

Where N = Speed in RPM, T = Number of teeth.

Module 4: Cams, Tappets and Variable Speed

Duration: 8 Hours | CO4

M4.01: Tappets and Cams

Cams and Tappets convert **Rotary Motion** into **Reciprocating (up and down) Motion**.

Textile Application: In a weaving loom, Tappet shedding uses cams to lift and lower heald frames to create the shed for the shuttle to pass through.

M4.02: Variable Speed Drives

Textile processes often require speed changes (e.g., as a bobbin builds up in size).

- **Stepless Cone Pulleys:** Used in the Roving frame to gradually change the speed of the bobbin as its diameter increases.
- **PIV (Positive Infinitely Variable) Drives:** A chain-based drive that allows precise, continuous adjustment of the output speed.

Formula Bank

1. Belt Drive Speed Ratio

$$N_2 / N_1 = D_1 / D_2$$

N: RPM, D: Diameter

2. Gear Speed Ratio

$$N_2 / N_1 = T_1 / T_2$$

T: Number of Teeth

3. Belt Slip Percentage

$$\% \text{ Slip} = [(V_1 - V_2) / V_1] \times 100$$

V₁: Surface speed of driver, V₂: Surface speed of driven

Interactive Speed Calculator

Driver Speed (N1) RPM:

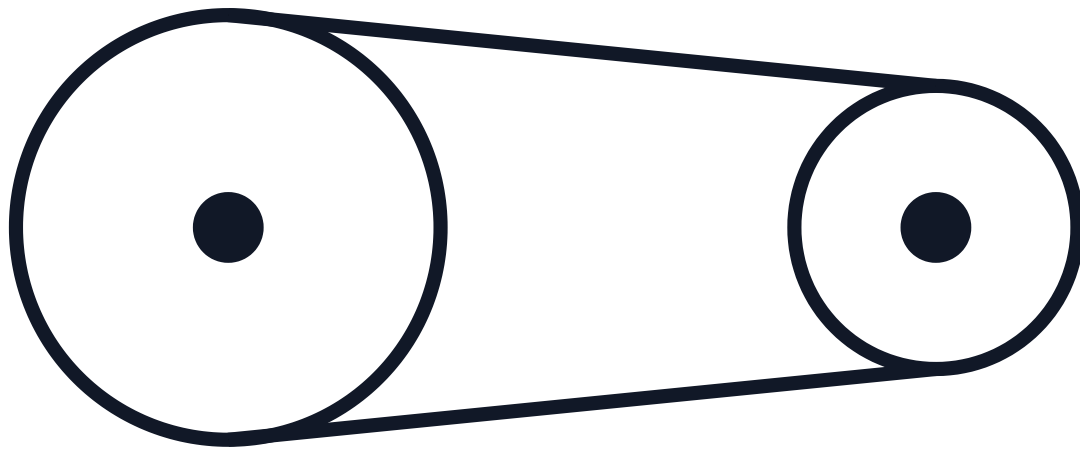
Driver Teeth/Dia (T1):

Driven Teeth/Dia (T2):

Resultant Speed (N2):

2880 RPM

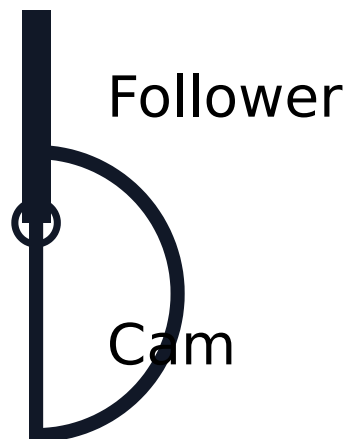
Diagram Library



Driver (D1)

Driven (D2)

Figure 1: Simple Open Belt Drive Schematic



Follower

Cam

Figure 2: Basic Cam and Follower Mechanism

Practice Q&A

1. What is the main difference between a Shaft and an Axle?

A shaft is designed to transmit torque and power (it rotates), whereas an axle primarily supports the weight of rotating components and may or may not rotate.

2. Why are V-belts preferred over flat belts for short distances?

V-belts provide better grip due to the wedging action in the pulley groove, allowing for higher torque transmission with less slip at short center distances.

3. What is a 'Positive Drive'?

A drive where there is no possibility of slip between the driver and driven members, ensuring a constant velocity ratio. Examples: Gear drives and Chain drives.

4. Define 'Slip' in belt drives.

Slip is the relative motion between the belt and the pulley surface, occurring when the frictional grip is insufficient. It reduces the velocity ratio.

Practice Model Lab Exam

Time: 3 Hours | Max Marks: 75 (Internal/External)

1. Identify the given mechanical elements (Shaft, Pulley, Key, Bearing) and write their functions. (10 Marks)
2. Sketch a compound gear train and calculate the final output speed given the driver speed and tooth counts. (20 Marks)
3. Demonstrate the working of a Tappet shedding mechanism on the allotted loom. (15 Marks)
4. Calculate the percentage of belt slip for the main motor drive of the carding machine. (15 Marks)
5. Viva-voce: Explain the application of PIV drives in textile machinery. (15 Marks)

Quick Revision

- **Mechanical Elements:** Shaft (Power), Axle (Support), Key (Fastener), Coupling (Connection).
- **Transmission:** Belts (Flexible/Slip), Gears (Rigid/Positive), Chains (Flexible/Positive).
- **Gears:** Spur (Parallel), Bevel (Intersecting), Worm (High Reduction).
- **Cams:** Convert Rotary to Reciprocating motion.
- **Variable Speed:** Cone pulleys and PIV drives are essential for winding processes.

Glossary

Torque

The twisting force that tends to cause rotation.

Velocity Ratio

The ratio of the speed of the driver to the speed of the driven member.

Shedding

The process of separating warp threads into two layers to form a tunnel (shed).

Heald Frame

A component in a loom that holds the heald wires through which warp ends pass.

Official Source Declaration

This content is strictly based on the **SITTTR Kerala Revision 2021** syllabus for **Course Code 2069: Basic Textile Mechanics Lab**. All modules, outcomes, and technical topics align with the official curriculum provided by the State Institute of Technical Teachers' Training & Research.

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- **Calculations:** 计算总牙数 (Total Teeth) 和平均牙数 (Average Teeth)。
总牙数 = 上颌牙数 + 下颌牙数。
平均牙数 = 总牙数 / 样本数。